



Carpooling Platform

1. Introduction

Problem Statement

Daily commuting is a significant challenge for employees, often leading to increased transportation costs, traffic congestion, fuel consumption, and environmental impact. Many employees travel along similar routes and schedules but lack an efficient way to coordinate shared transportation.

The objective of this hackathon is to develop an **Enterprise Carpooling Platform** that enables employees from registered organizations to discover and share rides. The platform should provide a seamless experience for finding rides, offering rides, managing trips, tracking journeys in real time, and handling payments while promoting cost-effective and sustainable commuting.

2. Solution Overview

The platform provides a complete ecosystem for ride sharing between employees.

Employees can use the application in two ways:

Find a Ride

Search for available rides by providing travel details such as pickup location, destination, date, and time. The system recommends matching rides that can be booked instantly.

Offer a Ride

Employees travelling in their own vehicles can publish rides by specifying their travel route, available seats, and fare.

Apart from ride matching, the platform also includes:

- Vehicle Management
 - Trip Management
 - Live Trip Tracking
 - Wallet & Payments
 - Ride History
 - Reports & Analytics
 - Company Administration
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3. User Roles

Company Administrator

The Company Administrator is responsible for managing organization-wide carpooling information and ensuring employee and vehicle records remain accurate.

Responsibilities

- Manage employee records.
- Manage registered vehicles and driver information.
- Configure organization-specific carpooling settings.
- Maintain fuel cost, travel cost, and other operational configurations.
- Monitor employee participation.
- Provide employees access to the carpooling platform.

Note: The administrator is responsible for platform configuration only and is not involved in day-to-day ride operations.

Employee

Employees are the primary users of the platform.

A single employee can **offer a ride** while driving or **find a ride** when travelling as a passenger. These are activities performed by the same user rather than separate user roles.

Employees can:

- Register and manage their profile.
- Register and manage vehicles.
- Search available rides.
- Publish rides.
- Book rides.
- View and manage trips.
- Track rides in real time.
- Communicate with other participants.
- Complete payments.
- View ride history and reports.

4. Application Workflow

The application follows the workflow below:

1. User launches the application.
2. User logs in or creates a new account.
3. User chooses to either **Find a Ride** or **Offer a Ride**.
4. The system displays the calculated route for confirmation.
5. Depending on the selected option:
 - Search and book an available ride.
 - Publish a new ride.
6. Booked rides appear under **My Trips**.
7. During the journey, both participants can track the trip in real time and communicate through chat or call.
8. After the trip is completed, payment is made.
9. The completed ride is stored in **Ride History** and contributes to the reporting dashboard.

5. Functional Modules

5.1 Authentication

The application starts with a splash screen followed by employee authentication.

Features

- Employee Login
- New User Registration

- Profile Creation

After successful authentication, users are redirected to the dashboard.

Screens

- Splash Screen
 - Login
 - Sign Up
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5.2 Find a Ride

This module enables employees to search for rides that match their travel requirements.

Required Information

- Pickup Location
- Destination
- Travel Date
- Travel Time
- Number of Seats
- Recurring Ride

The application calculates and displays the route for confirmation before searching for matching rides.

Each ride displays:

- Driver Details
- Route
- Departure Time
- Available Seats
- Fare Per Seat

Screens

- Find Ride
 - Route Confirmation
 - Available Rides
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5.3 Offer a Ride

Employees travelling in their own vehicles can publish rides.

Before publishing a ride, users must register at least one vehicle.

Required Information

- Pickup Location
- Destination
- Travel Date & Time
- Available Seats
- Fare Per Seat

Once the route is confirmed, the ride becomes available for booking.

Screens

- Offer Ride
 - Route Confirmation
 - My Vehicle
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5.4 Trip Management

After booking, rides appear in **My Trips**.

Trip Information

- Driver Details
- Passenger Details (Driver View)
- Vehicle Information
- Pickup & Drop Locations
- Trip Schedule
- Fare Details
- Current Trip Status

Trip Lifecycle

- Ride Booked
- Trip Started
- Trip In Progress
- Trip Completed
- Payment Pending
- Payment Completed

Communication

Passengers and drivers can communicate throughout the trip using:

- Chat
- Voice Call

These features help coordinate pickup locations, delays, and other trip-related updates.

5.5 Live Trip Tracking

Live Trip Tracking is a mandatory feature of the application.

Once the driver starts the trip, both participants can monitor the journey using an interactive map.

Tracking Information

- Live Vehicle Location
- Current Route
- Estimated Arrival Time (ETA)
- Pickup Marker
- Destination Marker
- Current Trip Status

Live tracking remains active until the trip is completed.

5.6 Payments & Wallet

Once a trip is completed, passengers can make payments using one of the supported methods.

Payment Methods

- Cash
- Card
- UPI
- Wallet

Wallet Features

- View Balance
- Recharge Wallet
- Pay Using Wallet Balance

Screens

- Payment
 - Wallet
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5.7 Ride History

Ride History maintains a record of all completed trips.

Each entry contains:

- Ride Participants
 - Route
 - Vehicle Information
 - Date & Time
 - Trip Status
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5.8 Vehicle Management

Drivers can register and manage multiple vehicles.

Vehicle Information

- Vehicle Model
- Registration Number
- Seating Capacity

Only registered vehicles can be selected while publishing rides.

5.9 Reports & Analytics

The reporting dashboard provides insights into travel activity and transportation costs.

Reports

- Total Trips
- Total Distance Travelled

- Fuel Consumption
 - Cost Per Kilometer
 - Vehicle-wise Cost Analysis
 - Fuel Efficiency Trends
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5.10 Settings

The Settings module provides quick access to commonly used features.

Available Options

- My Trips
- My Vehicle
- Payment Methods
- Ride History
- Saved Places
- Help & Support
- Chat

Saved Places

Users can save frequently used pickup and destination locations, such as **Home**, **Office**, or other preferred locations, making future ride searches and ride publishing faster.

6. Functional Requirements

The solution should implement the following capabilities.

User Management

- Employee Registration
- Employee Login
- Profile Management
- Company Administration

Ride Management

- Search Ride
- Publish Ride
- Route Confirmation
- Ride Matching
- Ride Booking
- Trip Management
- Live Trip Tracking

Vehicle Management

- Register Vehicle
- Update Vehicle Information
- Manage Seat Availability

Payment Management

- Cash Payment
- Card Payment
- UPI Payment
- Wallet Payment

Wallet Management

- Recharge Wallet
- View Wallet Balance
- Wallet-based Ride Payments

Reports & Analytics

- Ride History
 - Travel Reports
 - Cost Analysis
 - Fuel Consumption Reports
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7. Assumptions

The following assumptions apply to the hackathon implementation:

- The platform supports multiple organizations, each with its own registered users and administrator.
 - Only authenticated users belonging to a registered organization can access the platform.
 - Every ride has one driver and one or more passengers based on the available seating capacity.
 - Drivers must register at least one vehicle before publishing a ride.
 - Route generation and live trip tracking will use a mapping service (Google Maps, OpenStreetMap, or equivalent).
 - Live location sharing is enabled only while a trip is active.
 - Payments should be implemented using Razorpay Test Mode or an equivalent sandbox environment. Real money transactions are not required for the hackathon.
 - Reports are generated from trip, vehicle, and travel data collected by the application.
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8. Evaluation Expectations

Participants are free to choose any technology stack.

The solution should demonstrate the complete end-to-end business workflow described in this document.

Mandatory Features

- Authentication
- Ride Discovery
- Ride Publishing
- Route Confirmation
- Ride Booking
- Trip Management
- Live Trip Tracking
- Vehicle Management
- Payments & Wallet
- Ride History
- Reports Dashboard

Bonus Features

Participants are encouraged to implement additional capabilities such as:

- Ride Notifications
- Ride Cancellation
- Intelligent Ride Matching
- Route Optimization
- Enhanced Analytics
- Real-time Push Notifications

What Participants Will Learn

- Design and develop a real-world enterprise application.

- Build intuitive UI/UX and end-to-end business workflows.
 - Implement secure authentication and role-based access control.
 - Integrate third-party services like Maps and payment gateways.
 - Develop real-time features, dashboards, and analytical reports.
 - Design scalable system architecture and business logic.
 - Gain hands-on experience in full-stack application development.
 - Strengthen problem-solving, collaboration, and software engineering practices.
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Mockup:-

<https://link.excalidraw.com/I/65VNwvy7c4X/4OqWfsLBtnt>